

U.S. Multi Industry & Electrical Equipment

3M Co

Rating

Underperform

Price Target

MMM

131.00 USD



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3M: Is the R&D engine re-igniting? Thoughts on the EBO partnership with Microsoft for data center connectivity.

3M recently announced a **strategic partnership with Microsoft** centered around Expanded Beam Optical (EBO) technology - 3M's non-contact optical fiber connectivity technology for next-generation AI data centers. At its core, EBO addresses a key operational challenge in optical fiber connectivity: contamination from dust particles. Unlike traditional Multi-fiber Push On (MPO) connectors (which rely on physical fiber-to-fiber contact and require regular inspection and cleaning procedures), EBO uses a lens array to expand and refocus light across an air gap. The result is a connector that is inherently more tolerant to dust and contamination, while materially reducing installation complexity and deployment timelines. According to 3M, **installation times can be reduced from roughly three minutes (for a traditional MPO) to thirty seconds per connection (for EBO)**. There is some strategic significance beyond the Microsoft partnership. Generally, successful technology adoption begets more adoption, and validation by one large hyperscaler often accelerates broader industry uptake. We think Microsoft's endorsement serves as an important signal to other hyperscalers evaluating EBO optical fiber connectivity architectures for increasingly dense AI clusters, where reliability, deployment speed, and operational efficiency are critical design considerations. From a competitive standpoint, 3M appears reasonable well-positioned. With **patent protection extending until at least 2041**, participation in the broader EBO ecosystem through the Multi-Source Agreement, an early manufacturing lead, and the absence of credible hyperscaler-scale competitive offerings collectively strengthen the company's position. While important questions remain around monetization, pricing durability, and the potential emergence of alternative standards, we believe the announcement ([link](#)) represents a **meaningful milestone in the commercialization journey of EBO**.

For our own thesis on 3M, where we have highlighted weakness in R&D as a reason driver behind our tepid growth forecasts, **we see the partnership as an encouraging move in the right direction.**

Investment Implications

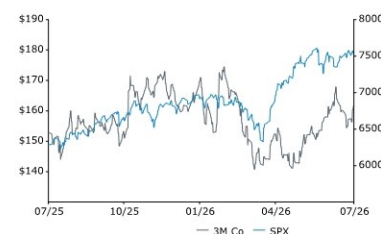
We rate MMM **Underperform** with a TP of **\$131**. We remain concerned longer-term around R&D and PFAS. However, we do acknowledge that this partnership indicates 3M is making progress in the right direction on R&D.

Adjusted EPS	F25A	F26E	F27E	Financials	F25A	F26E	F27E	CAGR	Valuation Metrics	F25A	F26E	F27E
MMM (USD)	8.06	8.53	9.32	Revenues (M)	24,948	24,951	25,641	--	Adjusted P/E (x)	20.1	19.0	17.4

Source: Bloomberg, Bernstein estimates and analysis.

Close Date	16 Jul 2026			
MMM Close Price (USD)	161.77			
Price Target (USD)	131.00			
Upside/(Downside)	(19)%			
52-Week Range	177.41/139.34			
SPX	7,533.77			
FYE	Dec			
Div Yield	1.9%			
Market Cap (USD) (M)	84,374			
EV (USD) (M)	93,401			
Performance	YTD	1M	6M	12M
Absolute (%)	1.0	1.6	(3.6)	1.7
SPX (%)	10.1	0.3	8.6	20.3
Relative (%)	(9.0)	1.3	(12.1)	(18.6)
Source: Bloomberg, Bernstein estimates and analysis.				

Price Performance, 1YR



DETAILS

Optical fiber cables are the circulatory system of modern data centers. Rather than transmitting electrical signals through copper wires, they carry information as pulses of light, enabling high bandwidth, low latency, and efficient transmission across the entire data center ecosystem. As AI rack densities continue to increase, the volume of data moving between GPUs, switches, and servers has increased exponentially, making optical fiber the preferred medium for high-performance networking and data transfer. In simple terms, the primary function of optical fiber is to move data quickly, reliably, and with minimal signal loss between servers, racks, rows and buildings, allowing thousands of servers to operate as a single coordinated system. In hyperscaler and AI-focused data centers, network performance is increasingly becoming as important as compute performance itself.

However, optical fibers are only useful if they can be efficiently connected together throughout the network. This is where optical fiber connectors come into play. Connectors are the physical interfaces that join one optical fiber to another, creating the data pathways that link servers, switches, storage systems, and networking equipment across a data center. Because light must pass from one fiber into the next with very high precision, connector quality directly impacts network reliability and performance. Traditional connector designs (called Multi-fiber Push On or MPO) rely on physical fiber-to-fiber contact, which makes them highly sensitive to contamination and handling.

What is the Expanded Beam Optical (EBO) technology?

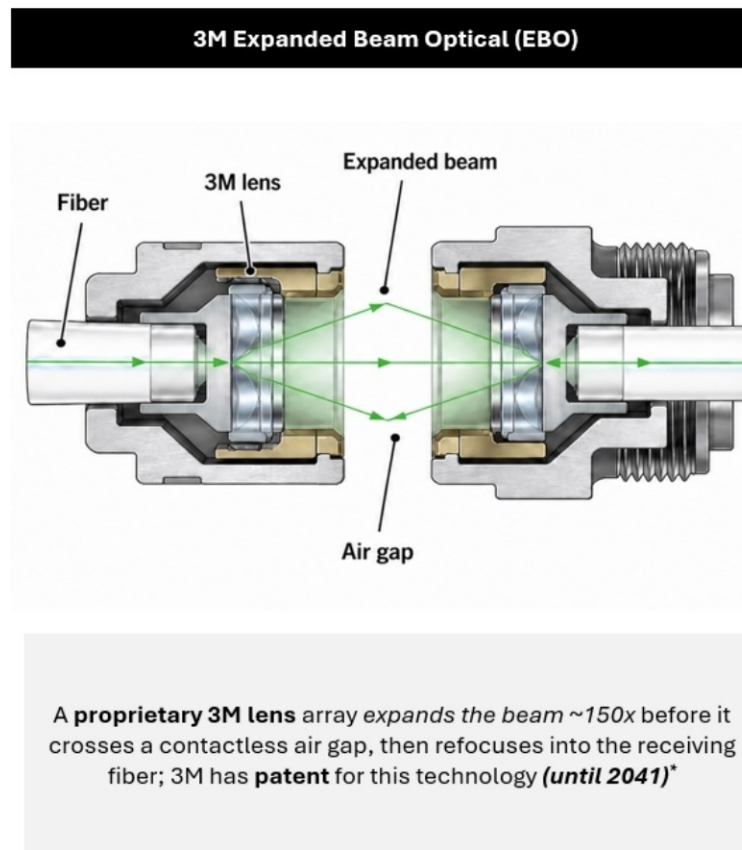
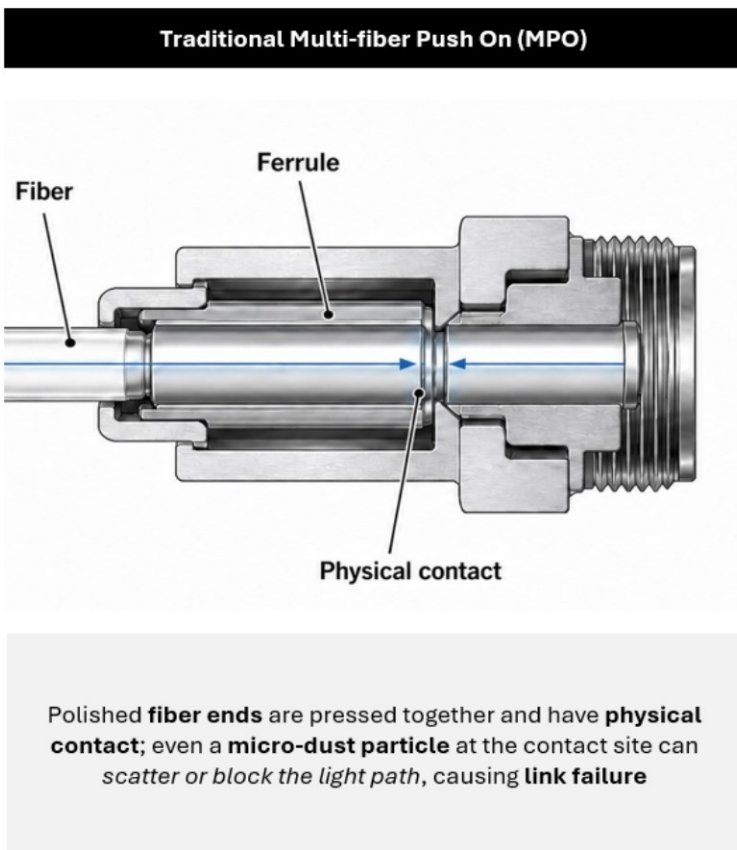
The fundamental innovation behind EBO is the elimination of physical contact between optical fibers. Traditional MPO connectors require fiber ends to be pressed together, making performance highly sensitive to contamination and creating a mandatory inspection-and-cleaning workflow prior to deployment. Even a micro-thickness dust particle can cause contamination that causes link failure. EBO replaces this architecture with a proprietary lens array that expands the optical beam significantly (to the order of ~150x) through a lens array before transmitting it across an air gap and refocusing it into the receiving fiber. By removing physical contact from the connection process altogether, EBO materially improves dust tolerance while simplifying installation procedures. 3M received a patent for EBO technology in Jan 2025, with an expiration through at least August 2041.

Advantages of EBO over Traditional MPO

The practical benefits of EBO become apparent when compared directly against traditional MPO connectors. The non-contact design materially reduces contamination sensitivity, removes inspection requirements, minimizes cleaning procedures, and shortens installation times by ~85% (according to a 3M whitepaper explaining deployment at a hyperscaler facility in the United States). Beyond the obvious labor savings, these improvements have meaningful implications for large-scale data center construction where thousands of optical fiber connections must be deployed and maintained.

Overview of 3M and Microsoft strategic partnership

In our view, the partnership creates clear value for both parties. For Microsoft, EBO offers lower installation costs, improved reliability in increasingly dense AI architectures, and faster deployment timelines that can accelerate the monetization of data center investments. Given the scale of hyperscaler capital expenditures, even modest improvements in deployment efficiency can translate into sizable economic benefits. For 3M, the value proposition is validation. Microsoft's adoption provides the first hyperscaler-scale validation of EBO, supports the rationale behind recent manufacturing expansion investments, and establishes an important customer reference point that could influence future adoption decisions across the broader hyperscaler data center ecosystem. Separately, Microsoft's Frontier Company initiative will support 3M's own enterprise transformation efforts, reinforcing management's ongoing focus on operational efficiency and process improvement.

EXHIBIT 1: **EBO** is an innovation in optical fiber-to-fiber “connectors”

← **EBO eliminates fiber-to-fiber physical contact**, removing mandatory pre-installation inspection & cleaning at each connection node (required in traditional MPO); **delivering inherently dust-tolerant performance** →

* United States patent publication number US2023/0056995A1

Source: Bernstein Analysis

EXHIBIT 2: **EBO is better than MPO across a range of technical attributes**

Theme	Traditional MPO	3M EBO
1 Connection mechanism	Direct fiber-to-fiber physical contact	No contact (beam crosses air gap via lens array)
2 Contamination sensitivity	High (one dust particle disrupts signal, causing overall link failure)	Low (beam is ~150x larger and much less sensitive as a result, air gap resistance is negligible)
3 Inspection requirement	Yes (at every connection node)	No (requirement eliminated entirely)
4 Cleaning requirement	Yes (at every connection node, at routine intervals)	Minimal
5 Installation time (per connector)*	~3 minutes	~30 seconds (~85% reduction in installation time)
6 Deployment time*	~6 months	~3 days

* Based on actual 3M deployment with a hyperscaler data center in the United States

Source: 3M Whitepaper (2025), Company Website, Bernstein Analysis

EXHIBIT 3: **Why the strategic partnership?**

What does 3M get?	What does Microsoft get?
1 First hyperscaler validation - Microsoft becoming the first Tier-1 cloud provider to deploy EBO transforms it from a promising pilot to production-validated technology at hyperscaler-level Sets a historical precedent: once one hyperscaler endorses, others follow within 12-24 months 2 Demand-led capacity expansion confirmed 3M announced >2x capacity expansion for US EBO manufacturing in March 2026 (4 months before this deal) - Microsoft deployment confirms that the supply-side bet was correct, de-risking the capex 3 Microsoft AI for internal enterprise transformation Microsoft Frontier Company (new enterprise AI vertical launched in July 2026) engineers deployed to automate 3M order management workflows (credit checks, delinquency, system updates etc.) Augments ongoing 3M operational efficiency improvement narrative for investors	1 Operational cost reduction ~85% reduction in installation time per connector (vs MPO) to ~30 seconds (EBO) - Elimination of inspection & cleaning at every connection node removes significant technician labor and tooling requirements 2 Improved reliability in high-density AI rack architectures - Any micro-dust particle can degrade cluster performance by causing link failure EBO dust-tolerant non-contact design directly tackles the core problem of dense cabling environments 3 Reduced time to deployment (accelerated time to revenue) Compressed full deployment timelines pulls forward cash flows in hyperscaler data centers (where each rack starts generating revenue the moment it goes live) - At ~\$75B+ annual capex on data centers, this can have a direct material P&L impact

Source: Company Website, Bernstein Analysis

3M competitive advantage

3M's competitive advantage appears to stem from a combination of intellectual property, ecosystem positioning, and manufacturing execution. The company's patented ferrule and lens-array design effectively positions 3M at the center of the EBO ecosystem, with partner companies building solutions around the core technology through the Multi-Source Agreement framework (MSA partners can be thought of as integrated cable manufactures / assemblers, all of whom use the proprietary 3M ferrule design as a base). This creates opportunities spanning both direct product sales and potential licensing economics. Importantly, 3M remains the only participant to demonstrate EBO at meaningful hyperscaler data center scale, and recent investments in manufacturing capacity further reinforce their first-mover advantage. Potential competitors to 3M EBO include established fiber manufacturers (such as Corning, CommScope) and legacy EBO players (TE Connectivity, Amphenol which have EBO for military and industrial applications). None of these players currently offer comparable hyperscaler-focused EBO solutions, suggesting that 3M retains a meaningful lead in commercialization, at least as of today.

Key factors to watch out for

Despite the positive implications of today's announcement, several important questions remain. First, investors may struggle to quantify near-term financial impacts given that 3M does not separately disclose data center-related revenues and has not provided explicit content metrics (in \$/MW terms) tied to data center deployments. Second, while the MSA framework supports ecosystem adoption, it could eventually contribute to price pressure on certain components over time. Third, broader hyperscaler adoption is not guaranteed, as alternative connector standards could emerge from competitors or be adopted by other cloud operators. Finally, incumbent connectivity suppliers maintain deep customer relationships and possess substantial technical resources, creating the possibility of competing solutions (that do not overlap with 3M's patents) in future years. While none of these risks alter our positive interpretation of today's announcement, they remain important factors to monitor as the EBO ecosystem develops further.

EXHIBIT 4: Why 3M leads, and why that appears likely to be the case going forward as well?

3M edge	Competitive landscape assessment														
Patented ferrule design - Lens array and ferrule design is patent-protected (<i>granted in Jan 2025, valid until at least Aug 2041</i>)* Every EBO optical fiber assembly (regardless of the actual assembler), runs on a 3M ferrule Hub-and-spoke licensing model via EBO MSA 3M co-founded the EBO Multi-Source Agreement with TE Connectivity, Amphenol, AMD, Cisco, Meta, Oracle, Arista, Molex among others as partners - Partners "build" cables around the 3M ferrule as base Licensing royalties and component supply agreements compound revenue opportunity beyond direct sales First to demonstrate & commercialize scale 3M is the only company (till date) to demonstrate EBO at multi-fiber hyperscaler data center volumes This has accelerated commercial availability Manufacturing head start 3M announced >2x capacity expansion for U.S. EBO production with new advanced equipment in March 2026	<table> <tr> <th>Factor</th><th>Assessment</th></tr> <tr> <td>Patent protection</td><td>Strong near-term moat</td></tr> <tr> <td>Materials science complexity</td><td>Vapor coating + lens array design are non-trivial to reverse-engineer</td></tr> <tr> <td>Precision manufacturing</td><td>High barrier (3M is scaling manufacturing capacity precisely because of this)</td></tr> <tr> <td>Legacy EBO players (TE Connectivity, Amphenol)</td><td>Active in legacy EBO applications (military/industrial); not in hyperscaler data center use-cases yet</td></tr> <tr> <td>Established fiber manufactures (Corning, CommScope)</td><td>Could theoretically enter; but do NOT have any competing EBO product in lineup yet</td></tr> <tr> <td>Timeline to credible rival</td><td>3–5 years minimum; 3M will deepen customer lock-in by then</td></tr> </table>	Factor	Assessment	Patent protection	Strong near-term moat	Materials science complexity	Vapor coating + lens array design are non-trivial to reverse-engineer	Precision manufacturing	High barrier (3M is scaling manufacturing capacity precisely because of this)	Legacy EBO players (TE Connectivity, Amphenol)	Active in legacy EBO applications (military/industrial); not in hyperscaler data center use-cases yet	Established fiber manufactures (Corning, CommScope)	Could theoretically enter; but do NOT have any competing EBO product in lineup yet	Timeline to credible rival	3–5 years minimum; 3M will deepen customer lock-in by then
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Source: Company Website, Bernstein Analysis

EXHIBIT 5: **While a positive development, there are some considerations to watch out for**

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1 EBO revenue contribution yet to be quantified

- 3M does not report data center connectivity as a separate revenue segment; impact on reported financials can be difficult to track near-term
- Current guidance does not explicitly state a \$/MW content saving for customers

2 Open MSA standards could eventually commoditize EBO ferrule pricing

- By contributing to open-source interoperable specifications, 3M enables participation by the broader manufacturer ecosystem
- Primary structural risk is regarding ferrule pricing – but over a 3–5-year horizon (assuming new product developments do not interfere with 3M's patent)

3 Competing hyperscaler standards selection

- AWS / Google Cloud / Meta could endorse a competing connector standard, which could lead to a bifurcation in large-scale adoption among multiple players
- We need to watch out for fiber connector procurement disclosures from hyperscalers

4 Innovation from existing traditional MPO vendors (Corning, CommScope)

- Both have deep long-tenured relationships with hyperscalers
- A credible non-contact competing fiber connection solution (even 2-3 years out), could fundamentally alter the narrative

Source: Bernstein Analysis

The authors would like to thank Sakansh Mittal for his help in preparing this report.

APPENDIX - FINANCIAL FORECASTS

3M Financial Summary	FY2021A	FY2022A	FY2023A	FY2024A	FY2025A	FY2026E	FY2027E	FY2028E	FY2029E	FY2030E
Fiscal year ends: Dec 31	12/31/2021	12/31/2022	12/31/2023	12/31/2024	12/31/2025	12/31/2026	12/31/2027	12/31/2028	12/31/2029	12/31/2030
Income Statement (M)	FY2021A	FY2022A	FY2023A	FY2024A	FY2025A	FY2026E	FY2027E	FY2028E	FY2029E	FY2030E
Safety and Industrial	-	-	-	10,961	11,384	11,926	12,320	12,651	12,994	13,352
Transportation and Electronics	-	-	-	8,380	8,272	7,657	7,888	8,125	8,369	8,620
Consumer	-	-	-	4,931	4,920	5,003	5,103	5,205	5,309	5,416
Others	-	-	-	303	372	364	330	338	347	356
Total revenue	35,355	34,229	32,681	24,575	24,948	24,951	25,641	26,319	27,020	27,743
Total Gross Profit	16,560	14,997	14,204	10,128	9,957	10,403	11,026	11,580	11,889	12,207
Safety and Industrial	-	-	-	2,527	2,894	3,177	3,408	3,576	3,673	3,774
Transportation and Electronics	-	-	-	1,722	1,728	1,764	1,950	2,066	2,128	2,192
Consumer	-	-	-	932	996	1,037	1,099	1,158	1,181	1,205
Others	-	-	-	818	1,071	1,052	1,074	1,131	1,154	1,177
Underlying EBIT	7,832	6,947	6,388	5,067	5,693	5,993	6,433	6,773	6,954	7,142
EBIT margin safety and industrial	-	-	-	23.1%	25.4%	26.6%	27.7%	28.3%	28.3%	28.3%
EBIT margin transportation and electronics	-	-	-	20.5%	20.9%	23.0%	24.7%	25.4%	25.4%	25.4%
EBIT margin consumer	-	-	-	18.9%	20.2%	20.7%	21.5%	22.2%	22.2%	22.2%
Group EBIT margin	22.2%	20.3%	19.5%	20.6%	22.8%	24.0%	25.1%	25.7%	25.7%	25.7%
Net income to shareholders	5,921	5,777	(6,995)	4,009	3,250	3,997	4,812	5,002	5,145	5,293
Diluted underlying EPS (\$)	10.73	9.87	9.24	7.30	8.06	8.53	9.32	9.91	10.28	10.58
Diluted shares outstanding (m)	585	568	554	552	541	533	525	513	509	509
Balance Sheet (M)	FY2021A	FY2022A	FY2023A	FY2024A	FY2025A	FY2026E	FY2027E	FY2028E	FY2029E	FY2030E
PP&E	9,429	9,178	7,690	7,388	7,101	6,965	6,830	6,741	6,693	6,680
Other non-current assets	22,240	31,767	34,201	23,984	21,346	20,909	20,616	20,396	20,238	20,134
Other current assets	10,839	14,688	16,379	15,884	16,387	16,771	18,811	20,982	25,083	29,269
Cash and cash equivalents	4,564	3,655	5,735	5,600	5,235	6,863	8,759	10,803	14,719	18,712
Total assets	47,072	46,455	50,580	39,868	37,733	37,679	39,426	41,377	45,322	49,403
Current liabilities ex-debt	7,728	7,585	12,350	9,337	7,925	7,258	7,289	7,312	7,385	7,460
Total debt	17,363	15,939	16,035	13,044	12,602	12,556	12,556	12,556	12,556	12,556
Pension	2,870	1,966	2,156	1,813	1,631	1,586	1,586	1,586	1,586	1,586
Other non-current liabilities	3,994	6,195	15,171	11,780	10,828	10,601	10,601	10,601	10,601	10,601
Minorities	71	48	61	52	45	57	69	81	93	105
Total equity ex-minorities	15,046	14,722	4,807	3,842	4,702	5,622	7,325	9,241	13,101	17,095
Total equity and liabilities	47,072	46,455	50,580	39,868	37,733	37,679	39,426	41,377	45,322	49,403
Net debt	12,799	12,284	10,300	7,444	7,367	5,693	3,797	1,753	(2,163)	(6,156)
Net debt/ EBITDA	1.3x	1.4x	1.2x	1.2x	1.1x	0.8x	0.5x	0.2x	-0.3x	-0.7x
Underlying ROIC	-	19%	23%	-35%	23%	20%	24%	25%	26%	27%
Cash Flow Statement (M)	FY2021A	FY2022A	FY2023A	FY2024A	FY2025A	FY2026E	FY2027E	FY2028E	FY2029E	FY2030E
Underlying EBITDA	9,747	8,778	8,375	6,431	7,001	7,182	7,751	8,045	8,193	8,356
Working capital	(507)	(623)	535	201	(51)	453	(113)	(103)	(113)	(117)
Net capex	(1,552)	(1,549)	(1,496)	(1,120)	(815)	(980)	(1,026)	(1,053)	(1,081)	(1,110)
Other cash movements	(7,758)	(7,515)	(5,136)	(5,845)	(6,454)	(5,028)	(5,944)	(6,181)	(5,818)	(5,923)
Change in cash and cash equivalents	(70)	(909)	2,278	(333)	(319)	1,628	668	708	1,180	1,206

Source: Company reports, Bernstein estimates and analysis.

BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	16 Jul 2026		TTM Rel. Perf.	Adjusted EPS				Adjusted P/E (x)		
			Closing Price	Price Target		Cur	2025A	2026E	2027E	2025A	2026E	2027E
MMM (3M)	U	USD	161.77	131.00	(18.6)%	USD	8.06	8.53	9.32	20.1	19.0	17.4
MSFT (Microsoft)	O	USD	401.10	646.00	(41.9)%	USD	13.64	16.85	19.16	29.4	23.8	20.9
SPX			7,533.77									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

Source: Bloomberg, Bernstein estimates and analysis.

DISCLOSURE APPENDIX

I. REQUIRED DISCLOSURES

References to "Bernstein" or the "Firm" in these disclosures relate to the following entities: Bernstein Institutional Services LLC (April 1, 2024 onwards), Sanford C. Bernstein & Co., LLC (pre April 1, 2024), Bernstein Autonomous LLP, BSG France S.A. (April 1, 2024 onwards), Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, Sanford C. Bernstein (Canada) Limited, Sanford C. Bernstein (India) Private Limited (SEBI registration no. INH000006378), Sanford C. Bernstein (Singapore) Private Limited, Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社) and analysts employed by Société Générale Africa Technologies & Services to produce Bernstein research under a Global Services Agreement in place between Bernstein and Société Générale.

Bernstein is part of a joint venture between Société Générale (SG) and AllianceBernstein, L.P. (AB). Unless specifically noted otherwise, for purposes of these disclosures, references to Bernstein's "affiliates" relate to both SG and AB and their respective affiliates.

VALUATION METHODOLOGY**3M Co**

We value 3M using a DCF, with a terminal growth rate of 2.5% and WACC of 9.16% on our forecasted cash flows. We assume present value of PFAS related cash outflows (net of insurance recoveries) of \$11.5B. This implies a target price of \$131.

Microsoft Corp

We value Microsoft shares based on our estimate for 12-month forward NOPLAT per share in 1 year's time (\$22.02), and apply a P/FE multiple of 29x. We then add back in the net cash per share, discounted at 15% to account for potential tax costs and other "friction" to repatriate all cash (\$7.6 per share), arriving at price target of \$646.

RISKS**3M Co**

Upside: (i) Sustained industrial and consumer cycle reversal, (ii) Breakthrough, patent protected NPIs in high-growth sectors (DCs), (iii) Legal resolution of PFAS comes quicker than expected

Microsoft Corp

Risks to our price target for MSFT include: a decreased demand for Azure PaaS/IaaS; increased Cloud cybersecurity concerns that impact Cloud demand; lower than forecast sales of Microsoft SaaS offerings; increased competitive pricing pressure on IaaS; increased competition in the Office market; decreased PC sales; increased pirating of the company's software, especially Windows and Office; increased functionality in the virtualization layer that limits the need for Windows server on VMs; decreased monetization of Microsoft's online offerings especially Bing; failure to increase international market share of Bing; lower than forecast sales for Xbox products and services; decreases in Windows sales.

RATINGS DEFINITIONS, BENCHMARKS AND DISTRIBUTION**EQUITY RATINGS DEFINITIONS****Bernstein brand**

The Bernstein brand rates stocks based on forecasts of relative performance for the next 12 months versus the S&P 500 for stocks listed on the U.S. and Canadian exchanges, versus the Bloomberg Europe Developed Markets Large and Mid Cap Price Return Index EUR (EDME) for stocks listed on the European exchanges and emerging markets exchanges outside of the Asia Pacific region, versus the Bloomberg Japan Large and Mid Cap Price Return Index USD (JPL) for stocks listed on the Japanese exchanges, and versus the Bloomberg Asia ex-Japan Large and Mid Cap Price Return Index (ASIAX) for stocks listed on the Asian (ex-Japan) exchanges - unless otherwise specified.

The Bernstein brand has three categories of ratings:

- Outperform: Stock will outpace the market index by more than 15 pp
- Market-Perform: Stock will perform in line with the market index to within +/- 15 pp
- Underperform: Stock will trail the performance of the market index by more than 15 pp

Coverage Suspended: Coverage of a company under the Bernstein research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Not Covered (NC) denotes companies that are not under coverage.

Bernstein brand stock ratings are based on a 12-month time horizon.

Autonomous brand – common stocks

The Autonomous brand rates common stocks as indicated below. As our benchmarks we use the Bloomberg Europe 600 Financials Price Return Index (E600BK) and Bloomberg Europe Dev Mkt Financials Large and Mid Cap Price Ret Index EUR (EDMFI) index for developed European banks and Payments, the Bloomberg Europe 600 Insurance Price Return Index (E600IN) for European insurers, the S&P 500 and S&P Financials for US banks and Payments coverage, S5LIFE for US Insurance, the S&P Insurance Select Industry (SPSIINS) for US Non-Life Insurers coverage, and the Bloomberg Emerging Markets Financials Large, Mid and Small Cap Price Return Index (EMLSF) for emerging market banks and insurers and Payments. Ratings are stated relative to the sector (not the market).

The Autonomous brand has three categories of common stock ratings:

- Outperform (OP): Stock will outpace the relevant index by more than 10 pp
- Neutral (N): Stock will perform in line with the market index to within +/- 10 pp
- Underperform (UP): Stock will trail the performance of the relevant index by more than 10 pp

Coverage Suspended: Coverage of a company under the Autonomous research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Those denoted as 'Feature' (e.g., Feature Outperform FOP, Feature Under Outperform FUP) are our core ideas.

Not Covered (NC) denotes companies that are not under coverage.

Autonomous brand common stock ratings are based on a 12-month time horizon.

Autonomous brand – preferred stocks

The Autonomous brand has three categories of preferred stock ratings:

- Outperform (OP): The total return of the preferred instrument is expected to outperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Neutral (N): The total return of the preferred instrument is expected to perform in line with preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- Underperform (UP): The total return of the preferred instrument is expected to underperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.

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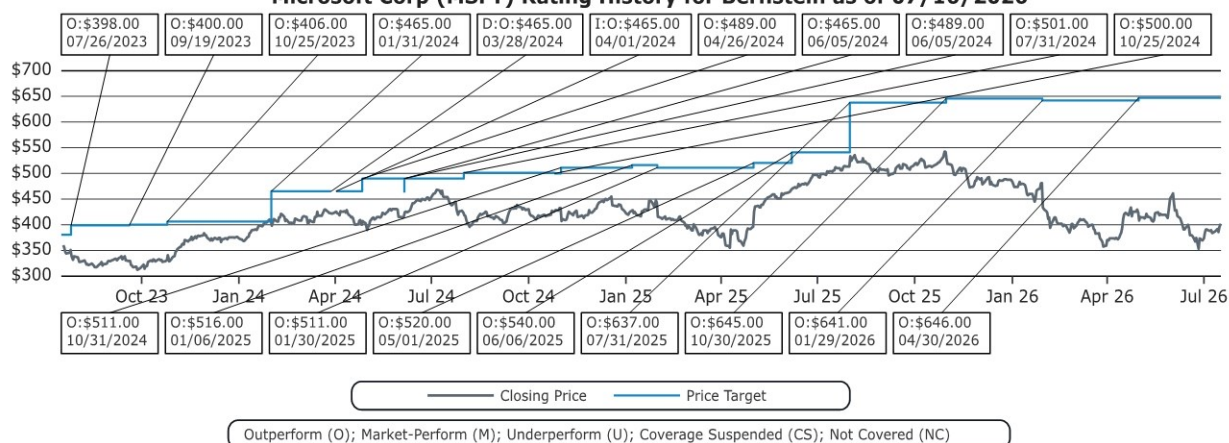
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